Comparative Analysis of Machine Learning Methods for Enhancing Intuitive Control of Upper-Limb Prosthetics

MREN 403 – Mechatronics & Robotics Design IV

Group 14 – Final Design Report

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Date Submitted: April 15th, 2026

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I. Introduction

Experiencing an upper extremity amputation is a traumatic and medically intensive undertaking. In countries with access to standard medical care there is a high rate of survival, with in-hospital mortality rates as low as 5% [1]. From there, patients must continue their lives with either reduced or no function of their upper limb(s). Prosthetics aim to return a level of normalcy to the individual, allowing them to regain confidence, independence, and employment. An alternative to prosthetics is vascularized composite allotransplantations (VCA), where a donor's upper extremity is transplanted onto the patient. When done properly, this best returns the lost abilities of the patient but poses a myriad of costs and potential complications [2]. For prosthetics there are 4 general categories; passive, body-powered, myoelectric, and hybrid (of body-powered and myoelectric) [3]. Each type presents advantages and disadvantages.

Myoelectric prosthetics offer a wide range of controls by using the local muscles' electric signals. To achieve more advanced control, the user is required to get Targeted Muscle Reinnervation (TMR). This is when the nerves that were connected to the amputated portion of the extremities get reinnervated to other muscles to increase the level of control [4]. This can be a large undertaking and presents challenges in cases where the patient has suffered a traumatic amputation. Furthermore, electromyography (EMG) sensors can be embedded in the user in conjunction with TMR to increase the level of control [5]. EMG signals are electric signals generated from contracting of muscles. The problem with myoelectric prosthetic arms is the control is not analogous to the control of a regular arm. There are multiple different control theories for them, which can be represented as: amplitude measurement, where signals are mapped into inputs either directly, proportionally, in a finite state machine or on/off control. And, feature extraction, where the features are run through pattern recognition or a regression analysis is done [6]. Essentially unique markers from muscle contractions actuate the arm by translating them to different motor controls [7].

Much research has been done in using electroencephalography (EEG) readings to control robotic prosthetics [8]. EEG signals are electric signals generated from brain activity. With the advancements in hardware, battery technologies, signal processing, and Artificial Intelligence, this area of prosthetics offers tremendous potential for improvement. By combining EEG and EMG signals to actuate a prosthetic, the user's intention can be interpreted with more accuracy [9], [10]. This will also increase the user's ability to control a prosthetic, as using EEG signals when the user has a higher level of amputation will help account for a lack of EMG sensors that can be used. Additionally, it can be expected that the improved accuracy allowed by using the EEG signals would negate requirements of

receiving TMR surgery. Allowing a prosthetic arm a wide range of control, through only non-invasive methods.

In this study we will train and evaluate models for the classification of motor movements through use of EEG and EMG signals. The purpose of which is to evaluate their use for intuitive control of prosthetics and finally evaluate the models using a signal to robotic arm pipeline. The purpose of which is for a more representative evaluation of the systems deployment on an edge device.

II. Methodology

The primary goal of this research is to evaluate the classification of EEG and EMG signals for human motor movements, for control of a representative prosthetic device. It can be hypothesized that EEG signals can be used to classify motor movements and more intuitive control can be achieved with EMG signals. We begin by outlining the hardware and software used for these evaluations. Next, we describe the dataset we used and generated, providing us with a large set of data to train models with and cross-validation of the models with our own data. Finally, we introduce the models used and the metrics for which they were trained and evaluated with.

A. Hardware

The robotic arm used to operate as a representative prosthetic was an Elephant Robotics myCobot 280 (Elephant Robotics, Shenzhen, China), shown in **Figure 1** below. The myCobot 280 is a prebuilt robotic arm with 6 degrees-of-freedom (DOF), with the ability to attach an adaptive gripper to the end-effector [11]. This provides 7-DOF, giving it similar functionality to a human arm [12]. It is controlled through serial communication (UART) from a PC using a USB to TTL serial label connected to an onboard ESP32 system on a chip. Joint angles and the gripper state are sent as commands from the control system. While this is not a true prosthetic, it provides a consistent and repeatable platform for testing how well the models are able to translate predicted movements into physical actions.

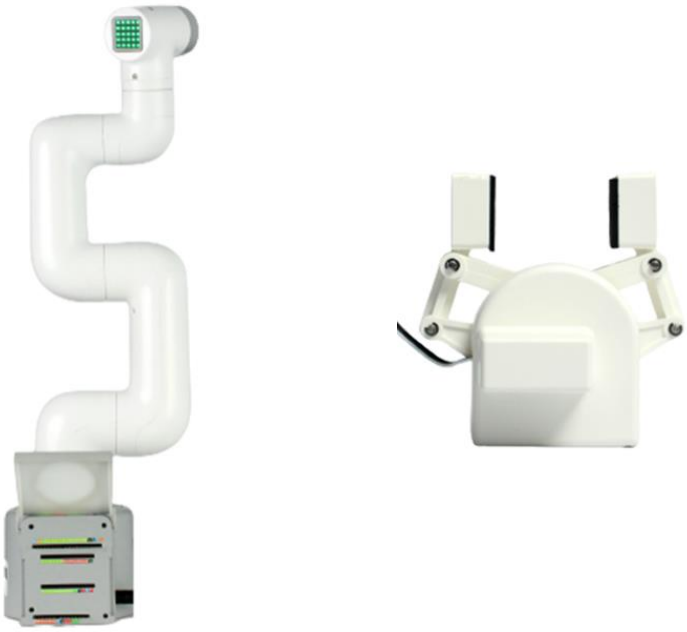

Figure 1: Elephant Robotics myCobot 280, with the adaptive gripper shown on the right [11].

B. Software

A ROS2 system architecture was used to control the robot while processing the EEG and EMG signals in parallel, on a PC. The system was programmed using Python and the myCobot GitHub library. The control provided was input to the joint angles in degrees and the state of the gripper. For the training of the models, the pipeline was programmed in Python using the PyTorch toolkit, all done on NVIDIA A100 GPUs through Google Colab. The EEG training models (EEG-TCNet, MSCFormer, EEGEncoder) were downloaded from their respective GitHubs. For system validation a graphical user interface (GUI) was made to function as a digital twin to the robotic arm. It was configured to allow the user to select movements to be inferred by the trained models and observe the results, the layout for which is shown in **Figure 5**, within **B. Additional Figures**.

C. Dataset

For the purposes of this study a large dataset with both EEG and EMG data was required. A dataset by Jeong et al. GigaScience 2020 [13], is used, in which EMG and EEG signals of 11 intuitive upper limb movements are recorded in unison. The experimental setup was designed to synchronously capture multimodal neurophysiological signals during intuitive upper extremity movements. Data was acquired from twenty-five healthy, right-handed participants over three separate recording sessions spaced one week apart. The hardware array consisted of a 60-channel electroencephalogram (EEG) positioned according to standard international specifications, a 7-channel electromyogram (EMG) targeting

specific right arm muscles to record physical activity and confirm motor stillness during cognitive tasks, and a 4-channel electro-oculogram (EOG) isolated for ocular artifact removal. The experimental paradigm required participants to perform both the physical execution and the motor imagery of eleven distinct tasks using a single upper limb. These tasks were categorized into three primary functional movements: multi-directional arm-reaching across six spatial directions, hand-grasping of three different objects, and wrist-twisting utilizing two distinct motions. Providing 11 movements in total, with the addition of a rest classes using the before and after windows from the movement execution. Each trial was timed using a visual cue protocol, providing participants with a four-second window to either execute or imagine the specified movement. For our purposes only the real movements data was used. For the models the train-test split of the subject-independent cases used 24 subjects for training and 1 for testing.

For additional validation, our team gathered EEG data from a single subject. Using a 32-electrode cap, due to resource constraints, the experimental paradigm was matched. Only 8 of the 11 movements were captured for approximately 20 trials each, again due to resource constraints. Of these 32 electrodes, 27 were used for future training from both our data and the dataset used, as some of the 32 electrodes were those used for EOG in the dataset. This data was only used for validation and not for training.

D. Models

The models used for EEG classification were selected to provide variability between their respective strengths and weaknesses. Additionally, their respective performance on a BCI Competition dataset was used as a benchmark. The BCI Competition dataset was created to validate signal processing and classification methods for Brain-Computer Interfaces (BCIs) [14] created by the Berlin Brain-Computer Interface. Due to the long-term considerations of deployment on an edge device, the trainable parameter counts and number of Million Multiply-Accumulates (MMACs) per inference was used as a metric for model selection. MMACs per inference is a measure of the total number of operations a network performs to process a single input. All models, with their respective parameter counts and MMACs per inference are shown in **Table 1**.

The first chosen model was EEG-TCNet created by Ingolfsson et al. in 2020, a temporal convolutional network (TCN) with a low trainable parameter count of 4272 and 6.8 MMACs per inference. For the 4 class BCI Competition IV-2a dataset it achieved 77.35% classification accuracy, which was further improved for individual subjects to 83.84% [15]. The second model used was MSCFormer by Zhao et al. in 2025, a Multi-Scale Convolutional Transformer (MSCFormer) that excels in generalization. It has the highest parameter count of all models used, at 236 220 and MMACs per inference at 90.26, while

achieving the highest subject-independent accuracy using BCI IV-2a, at 82.95% [16]. The last EEG classification model used was EEGEncoder created by Liao in 2024, which uses multiple parallel Dual-Stream Temporal-Spatial (DSTS) structures consisting of a TCN and stable transformer feeding into a multi-layer perceptron (MLP). The benefit of this structure is robustness to noise and motion artifacts. Its subject-independent accuracy was 74.48%, and 86.46% for subject-dependent, using the BCI IV-2a dataset. While having 180 000 trainable parameters and 22.5 MMACs per inference. All EEG classifications architectures (EEG-TCNet, MSCFormer, EEGEncoder) are shown in **B. Additional Figures**, in **Figure 2**, **Figure 3**, and **Figure 4** respectively.

Table 1: EEG classification models used, with their respective number of trainable parameters, and MMACs per inference, and performance using the BCI IV-2a dataset for both subject independent and dependent [15], [16], [17].

Model	EEG-TCNet	MSCFormer	EEGEncoder
Trainable Parameters	4 272	236 220	180 000
MMACs per inference	6.8	90.26	22.5
BCI IV-2a Accuracy Subject-independent	77.35%	82.95%	74.48%
BCI IV-2a Accuracy Subject-dependent	83.84%	–	86.46%

For classification of the EMG signals, various custom architectures were created and trialed, based on the current literature on classification of motor movements through surface EMG (sEMG) signals. The accuracy of the researched models was reported using dataset from the Ninapro database, which is focused on machine learning research on prosthetic hands [18]. These models featured both regression and classification heads; the purpose of the regression head was for a more novel control approach, and the classification head was to aid in optimization of the regression head. The style of regression used was multivariate, the final linear projection layer maps the hidden state to a 7-dimensional vector, representing the 6 joint angles and state of the gripper.

All models used a Tanh (Hyperbolic Tangent) activation function at the end of the regression head. The purpose of this is to restrict regression outputs into a range between -1.0 and 1.0, to build robustness to outliers and to ensure the neural network mathematically cannot issue commands that exceed the desired paths of the robot.

The created models are listed in **Table 2**, with their respective parameter counts and MMACs. The purpose of creating custom architecture was for a novel approach to myoelectric control. All models take a small 200 ms window, with a 50 ms stride, of the 4 second signal and infers it into the intended joint space of robotic arm at that time. By

creating robot paths through the joint angles and gripper state for each of the 11 intuitive movements, the model's regression head was trained to predict those values at that time in the signal.

The first custom architecture used was a transformer using an adaptive average pooling layer to compress the temporal dimension, with 87 795 trainable parameters and 12.8 MMACs. Following a similar conceptual approach to the dimensionality reduction techniques seen in LST-EMG-Net, created by Zhang et al. (2023), who achieved an accuracy of 81.47% on Ninapro DB2E2 [19].

The second created model was a multi-scale 1-dimensional Convolutional Neural Network (CNN), using a similar style model to that of Emimal M et al. (2023). Who achieved classification accuracy of 92.8% for Ninapro DB2 [20]. This model employs multiple parallel convolutional branches with varying kernel sizes. The purpose of this is to allow the network to capture high-frequency fluctuations and low-frequency patterns within the muscle activation window. This model had 129 395 trainable parameters and a MMACs per inference of 1.95.

Then using a vision-based approach, the 1-dimensional time-series signal was transformed into a 2-dimensional time-frequency signal (Spectrogram) using a Short-Time Fourier Transform (STFT). The intention of this was to evaluate the signals with a neural network that has established success in a different style classification problem. The spectrograms for each EMG electrode channel were fed into a ResNet-18 architecture customized to take 6-channels in place of the standard RGB layers, with a global average pooling layer at the end to convert into classification and regression predictions. The trainable parameters of this configuration were 11 195 667 and a MMACs per inference of 83.2. In 2022, Xu et al. had success using a 50-layer CNN based on residual networks (ResNet) to classify the Ninapro DB2 dataset with an accuracy of 99.51% [21]. Our design uses less layers to help reduce the computational load for future deployment on an edge device.

Lastly, a Multi-Head TCN Long Short-Term Memory (LSTM) was created, using a similar strategy to that of the multi-scale 1-D CNN. Parallel TCN branches with varying kernel sizes were used to capture features at varying frequencies before being fed into two LSTM to aid in smoothing the trajectory. This was following the methodology of Du et al. (2024) where a similar model was made for continuous estimation of wrist joint angles. They achieved an average root mean square error (RMSE) of 0.064 for their regression task. Shin et al. achieved an accuracy of 94.31% on the Ninapro DB1 dataset in 2024, with a similar style TCN-LSTM model focused on classification [22].

Table 2: The trainable parameters and MMACs per inference of the EMG classification models.

Model	Adaptive Transformer	Multi-Scale 1D-CNN	ResNet-18 Spectrogram	Multi-Head TCN-LSTM
Trainable Parameters	87 795	129 395	11 195 667	254 195
MMACs per inference	12.8	1.95	83.2	26.4

E. Training Parameters and Validation Metrics

The original architectures for TCNet, MSCFormer, and EEGEncoder were benchmarked on the BCI Competition IV dataset, which is balanced across all movement classes. Consequently, the original implementations relied on a standard Cross-Entropy loss function shown in (1), where *M* represents the total number of discrete classes, *y* is a binary indicator determining if class *c* is the correct classification, and *p* is the predicted probability of class *c*.

$$L_{cls} = -\sum_{c=1}^{M} y_{o,c} \log(p_{o,c}) \qquad (1)$$

However, the experimental protocol for the 11-intuitive movements dataset inherently created a skewed distribution among the classes. Since each trial consisted of a 4 second rest period, a 3 second visual cue pause, and a 4 second motor movement execution, this produced a disproportionate volume of rest trails relative to the other actions performed. Furthermore, certain gestures were consolidated to improve accuracy, notably supination was combined with pronation to form one wrist-twisting class, and the three distinct grasping variations were combined as one overall grasping class. These changes altered the amounts of overall samples, further skewing the class distributions. The classes were mathematically balanced to preserve the amount of data available, using a Class-Weighted Focal Loss with label smoothing. Shown in (2), a modification of the standard Cross-Entropy loss is done by introducing two critical parameters, a balanced class weighting (α_t), and a focusing parameter (γ). The optimal focusing parameter and label smoothing factor for the 11 intuitive movements dataset were found to be $\gamma = 3.0$ and 0.05, respectively. The balanced class weighting acts as a scalar for each class and is weighted inversely proportional to class frequencies, ensuring classes with fewer parameters have a larger impact on the loss.

$$FL(p_t) = -\alpha_t (1 - p_t)^{\gamma} \log(p_t) \qquad (2)$$

Finally, label smoothing was introduced, forcing the model to predict percent certainties for each class instead of predicting one class entirely. This parameter acts as a regularizer, which helps to mitigate overfitting.

Data augmentation was used to address the scarcity of distinct movement trials and improve model generalizability. The Mixup method was used; it adds the scaled voltages of two distinct EEG trials together to form a new, merged segment. It then creates soft labels indicating what the modified segment is made of, e.g. 80% hand up, and 20% hand down.

A 5-fold cross-validation protocol was used during subject dependent training, to maximize the utility of each subject's training data. The data was initially partitioned by isolating 10% for final testing and 90% for training and cross-validation. During this phase, the 90% pool was iteratively split, using a different 20% slice for validation in each of the five folds. While these training loops were capped at a maximum of 1000 epochs, an early stopping mechanism was used to monitor validation loss. The epoch counts from all five folds were then averaged to determine the optimal training duration to be used in the final production run, where the model was trained on the entire 90% data pool. While subject-independent training used a patience window of 5 epochs and did not employ the 5-fold strategy, all other training parameters were kept consistent with those used in subject-dependent training. The subject independent model was trained without a 5-fold validation split and instead, a final model was trained directly with validation and 5 epoch early stopping.

Model performance was assessed using Accuracy, Precision, Recall, F1-score, and Cohen's Kappa metrics. Although accuracy provides a general measure of correctness, it can be misleading in the presence of substantial class imbalance, as Saito and Rehmsmeier (2015) showed, standard accuracy may present an overly optimistic view of performance on imbalanced datasets. To address this limitation, greater emphasis should be placed on the Precision, Recall, and F1-score metrics, which provide a more aggregated view of class-wise performance [23]. Precision is shown in (3), where TP and FP correspond to true positive and false positive, respectively. This metric measures how often the predicted actions are correct. Recall is shown as (4), where TP and FN correspond to true positive and false negative, respectively. This metric measures how many actions the model accurately predicted out of all the samples within its class. These two metrics are averaged to produce the F1-score, as seen in (5). Following the evaluation framework for multi-class and imbalanced classification discussed by Sokolova and Lapalme (2009), the F1-score is important in this training because it reflects performance across all movement classes rather than allowing larger classes to dominate the evaluation [24]. Finally, Cohen's Kappa, shown as (6) where P_o is the observed accuracy

and P_e is expected accuracy by chance. As noted by Schlögl et al. (2007), Kappa has been widely used in brain-computer interfaces (BCI) evaluation as it accounts for chance agreement and is less affected by class imbalance than raw accuracy [25].

$$Precision = \frac{TP}{TP + FP} \tag{3}$$

$$Recall = \frac{TP}{TP + FN} \tag{4}$$

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{5}$$

$$Kappa = \frac{P_o - P_e}{1 - P_e} \tag{6}$$

The proposed EMG architectures utilize a multi-task learning (MTL) framework, integrating classification and regression loss metrics. The purpose of this is to account for stability issues that would be present in a regression only model. MTL framework mitigates stability issues through three mechanism. First decoupling the movement intent from the signal magnitude, by dedicating the classification head to identifying the overall movement, while the regression head fine-tunes the joint commands. The second mechanism is multi-task regularization, for just regression the signal amplitude is tracked, whereas the classification objective will capture topological differences in the signals. Lastly to negate the flaw of a pure regression model where continuous control will predict a continuous value, even during rest, a state gatekeeper was used. During inference if the softmax probability of the rest class exceeds 80% confidence threshold, the regression outputs are masked to keep the robotic arm still.

To optimize the shared backbone and the dual output pathways concurrently, the network relies on two distinct objective functions evaluated over the training distributions. The classification head uses Cross-Entropy Loss, same as the EEG models, measuring the divergence between the model's predicted probability distribution and the ground-truth label, as shown in (1). The regression head utilizes Huber Loss, to evaluate the continuous joint kinematics across the 7-degrees of freedom. Shown in (7), where a is the residual difference between true and predicted joint value, δ is the transition threshold. For minor deviations, the loss curves quadratically to fine-tune exact joint angles. For larger deviations the penalty becomes linear to prevent data outliers from disproportionately skewing the model.

$$L_{reg}(a) = \begin{cases} \frac{1}{2}a^2 & for\ |a| \leq \delta \\ \delta\left(|a| = \frac{1}{2}\delta\right) & for\ |a| > \delta \end{cases} \quad (7)$$

During backpropagation the gradients from both pathways are used to update the shared feature-extraction layers. The final scalar loss calculation is shown in (8), where the balancing coefficient, λ, is set to 100.0. As the classification loss is calculated across multiple probability classes and the regression loss is bounded by Tanh, the balancing coefficient prevents the classification head from dominating the training.

$$L_{total} = L_{cls} + \left(\lambda \times L_{reg}\right) \quad (8)$$

III. Results

A. EEG Classification

Initial evaluation of EEG classification performance was conducted before any classes were merged. Across the three models moderate accuracy was observed as seen in **Table 3**, with notably low precision, recall, and F1-scores, indicating strong class confusion and poor distinction between similar movements. As previously stated, closely related classes were merged and then used for training. The consolidation resulted in a measurable increase in overall classification performance across all models, particularly in F1-score and Kappa, reflecting more consistent and reliable predictions. These improved results can be found in **Table 4**.

Table 3: Evaluated metrics for models trained and tested on subject 1, before classes were amalgamated.

Subject	Metric	EEG-TCNet	MSCFormer	EEGEncoder
1 Male 11 classes	Accuracy	63.44%	65.26%	72.81%
	Precision	42.97%	45.98%	57.54%
	Recall	49.45%	50.26%	62.63%
	F1-Score	44.74%	47.04%	58.78%
	Kappa	52.10%	54.47%	63.98%

Table 4: Evaluated metrics for models trained and tested on subject 1 after combining classes

Subject	Metric	EEG-TCNet	MSCFormer	EEGEncoder
1 Male 9 classes	Accuracy	70.69%	74.32%	76.43%
	Precision	52.57%	51.44%	57.82%

	Recall	59.51%	57.92%	61.40%
	F1-Score	54.56%	53.22%	58.67%
	Kappa	61.51%	65.54%	68.10%

Following the improvements observed after class consolidation, additional subject-dependent models were trained across four subjects. These results are in **Table 12** in **C. Additional Results**. While model performance remained relatively consistent across architectures, classification metrics varied more significantly between subjects, indicating that subject-specific EEG characteristics have a greater impact on performance than the choice of model. The greatest variance in accuracy between subjects for the same model spanned a 7.88% range. In comparison to this, the largest variance between models for the same subject was only 5.74%. This suggests that the primary limitation in EEG-based classification may be from a documented phenomenon called BCI illiteracy, where certain individuals exhibit difficulty generating consistent and distinguishable brain signals for control tasks [26]. This leads to reduced model performance despite identical training conditions.

For the subject-independent approach results showed a significant drop in performance across all models. Accuracy decreased from approximately 70–76% in the subject-dependent case to as low as 35–54% as seen in **Table 5**. This highlights the difficulty of generalizing EEG patterns across different users. To evaluate if the results could be improved using all the EEG nodes from the dataset, models were also trained using the full 60-node configuration, the results, also shown in **Table 5**, showed some improvements in accuracy. It was also observed that training convergence occurred rapidly, with most models reaching their lowest validation loss within approximately 4 epochs.

Table 5: Evaluated accuracy of models training on subjects 2 through 25 and tested on subject 1.

Number of Nodes	Metric	EEG-TCNet	MSCFormer	EEGEncoder
27 Nodes	Accuracy	35.30%	48.26%	54.04%
	Precision	15.96%	20.92%	28.18%
	Recall	17.73%	15.56%	18.94%
	F1-Score	14.50%	12.28%	19.27%
	Kappa	15.42%	10.70%	21.21%
60 Nodes	Accuracy	46.19%	49.98%	50.77%
	Precision	14.10%	5.56%	24.32%
	Recall	16.11%	11.11%	15.95%
	F1-Score	12.59%	7.41%	14.22%
	Kappa	15.66%	0.02%	14.49%

The final evaluation done used our generated data with the 27-node trained models, as shown in **Table 6**. All models defaulted to predicting the same action for any input, which resulted in extremely poor performance. EEGEncoder achieved 0% accuracy as all predictions were for the Arm-reaching downwards class and while this trial was not collected. Across all models, prediction probabilities were in the range of 15% to 16% indicating that the models had low prediction confidence. This significant drop in performance is likely due to difficulties in replicating the controlled conditions of the original dataset. Some major differences in our collected dataset environment included bright lighting, background noise, and lack of a static room for data gathering.

Table 6: Number of correct guesses per trial using our collected data.

Action	Trials	TCNet	EEGEncoder	MSCFormer
Arm-reaching: Forward	19	0	0	0
Arm-reaching: Backward	21	0	0	0
Arm-reaching: Left	23	0	0	23
Arm-reaching: Right	19	19	0	0
Arm-reaching: Up	18	0	0	0
Arm-reaching: Down	0	0	0	0
Hand-grasping: Cup	17	0	0	0
Wrist-twisting: Pronation	38	0	0	0
Rest	149	0	0	0
Accuracy excluding rest (%)	NA	12.26%	0.00%	14.84%

B. EMG Classification

For the regression training the robot paths were used as the ground truths for the values during each window. The robots' paths were designed to attempt to mimic the complexity of each movement, where a more complex movement would use more joints, and a less complex movement would use less. For example, arm reaching forward vs. wrist supination. In **Table 11**, within **C. Additional Results** the robots' joint paths are shown. For the 4 second window the robot's joints would reach those values over the first 2 seconds then return to its rest position over the following 2 seconds.

The initial training was done using all 6 nodes, with no feature extraction or scaling methods used. The train-test split, as mentioned before, was subject-independent, training on 24 subjects and testing on 1. Initial results are listed in **Table 7**, note the Spectrogram classification and regression loss results are not available due to rapid overfitting of the model causing the training to stop.

Table 7: Initial validation results for all proposed models, showing accuracy with measured Cross-Entropy and Huber loss.

Metric	Adaptive Transformer	Multi-Scale 1D-CNN	ResNet-18 Spectrogram	Multi-Head TCN-LSTM
Accuracy	10.54%	28.71%	13.01%	32.05%
Cross-Entropy Loss	2.6682	2.0516	-	2.0058
Huber Loss	0.0166	0.0137	-	0.0135

As the multi-head TCN-LSTM model had the strongest performance it was used on the varying node levels. Feature extraction and scaling methods were evaluated, and the best results were achieved using zscore scaling and no feature extraction. The results are shown in **Table 8** for a randomly selected subject, and in **Table 9** for subject-independent. Overall, the single subject showed better results for the classification loss but, no overall measurable improvements. When comparing the subject-independent results across varying levels of nodes, the change from 6 to 4 nodes had a much smaller reduction in accuracy, near 3%. Whereas the change from 4 to 2 nodes reduced the accuracy by about 8%.

Table 8: Validation results for subject-dependent training using multi-head TCN-LSTM.

Nodes	6
Accuracy	37.66%
Cross-Entropy Loss	1.7142
Huber Loss	0.1554

Table 9: Validation results for subject-independent training using multi-head TCN-LSTM, across varying nodes.

Nodes	6	4	2
Accuracy	30.97%	27.79%	19.92%
Cross-Entropy Loss	2.0037	2.0652	2.2880
Huber Loss	0.0135	0.0140	0.0156

Due to these poor results and to test the validity of the best performing model, it was retrained using the three most discernable classes. Using a confusion matrix generated during training these classes were found to be Arm-reaching; backwards and leftwards, and Hand-grasping a ball. The results for which can be seen in **Table 10**. This showed a significantly improved accuracy, following the same reduction in node to accuracy relationship seen in the results when using all classes.

Table 10: Validation results for subject-independent training using multi-head TCN-LSTM, across varying nodes with the three most distinct classes.

Nodes	6	4	2
Accuracy	72.81%	71.30%	55.39%

Cross-Entropy Loss	0.6491	0.6924	0.9401
Huber Loss	0.0105	0.0106	0.0146

C. Robotic Arm Control

When evaluating the results on the myCobot 280, both the EEG and EMG models functioned as required with no noticeable delay. A problem arose when evaluating the EMG data with the regression outputs mapped to the joints. Although the Huber Loss would suggest that the joints would be within 4 to 9 degrees of their intended positions, in actuality, for a random movement the joints that were not required to move maintained their positions with ease whereas the joints that were required to move did not adhere to the expected 4 to 9 degrees of error. This is caused by the training easily creating regression outputs that predicted near zero values but severely struggled with properly mapping the moving joints.

IV. Discussion

The EEG classification portion of this project evaluated the performance of three deep learning architectures, EEG-TCNet, MSCFormer, and EEGEncoder, for classifying 12 distinct classes of EEG motor movement signals. The results show that generalizing neural patterns across subjects remains a major challenge. In the subject-dependent setting, the models achieved moderate performance, whereas subject-independent training led to a substantial drop in accuracy, with results ranging from 35.30% to 54.04%. This contrast suggests that inter-subject variability has a much greater effect on performance than model architecture alone. In other words, while the models can learn subject-specific EEG patterns reasonably well, those learned representations do not transfer reliably across different individuals. This outcome is also consistent with the concept of BCI illiteracy, which may further limit performance by making it difficult for some users to produce stable and distinguishable brain signals even under the same training conditions.

Despite EEGEncoder achieving slightly higher peak accuracy in isolated tests, EEG-TCNet is proposed as the optimal model for this system. The minor trade-off in accuracy is vastly outweighed by TCNet's lightweight architecture. For edge device implementation within a prosthetic, where computation, memory, and power are strictly constrained, a less computationally intensive model like TCNet ensures faster inference times and lower power consumption without sacrificing the reliability needed for practical applications.

Finally, the severe drop in performance observed from the evaluation of our generated data further demonstrates how sensitive EEG signals are to environmental noise, such as ambient lighting, background conversation, and motion artifacts. To address both these

environmental challenges and the limitations of BCI illiteracy, a transfer learning approach is highly recommended for future iterations. More specifically, future work should focus on pre-training a generalized EEG foundation model on a large and diverse dataset to learn robust baseline representations, then fine-tuning that model to individual subjects. This is supported by prior work showing that broadly pretrained EEG models can transfer across different subjects and hardware setups, and by more recent motor-imagery research demonstrating that fine-tuned transfer learning improved subject-independent classification accuracy from 72.22% to 79.44% on the BCI Competition IV-2a dataset, while also reducing inter-subject variability [27] [28].

The results from the EMG portion showed that the use of electrodes on the extensor digitorium and extensor carpi radialis brevis did little to improve the model's accuracy. Overall, the poor results were caused by the overt similarity of the movements, the signals produced were not discernable enough from each other, especially in regard to the EMG electrode configuration. Furthermore, the novel regression method is not currently a feasible way to refine the control of myoelectric prosthetics, if functional, the level of control would be improved. For future research on EMG models stochastic parameter regression would be employed, because of the way humans move, each movement has a discrete and continuous portion. The discrete intention to complete a movement and the continuous control of the force regulation and kinematics.

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Interests, Contributions, and Acknowledgements

Declaration of competing interests

The authors declare that they have no known competing financial interest or personal relationships that could have appeared to influence the work reported in this paper.

Authorship contribution statement

Leo Branigan: Conceptualization, Methodology, Writing – original draft.

Thomas Wilkinson: Conceptualization, Methodology, Writing – original draft.

Marco Shapria: Conceptualization, Methodology, Writing – original draft.

Ben Malcom: Conceptualization, Methodology, Writing – original draft.

Acknowledgments

The authors acknowledge the support for the conceptualization and methodology of this research from their supervisor Dr. Xian Wang. The support of Dr. Gerome Manson, his graduate students, and the Queen’s University Sensorimotor Exploration for the gathering of EEG data. And the support of Ph.D. candidate Graziella Bedenik for methodology and project supervision.

Appendix

A. Supplementary Data

Supplementary data to this article can be found online at
https://github.com/MarcoSchapira/EEG-Machine-Learning-Model-Comparison

B. Additional Figures

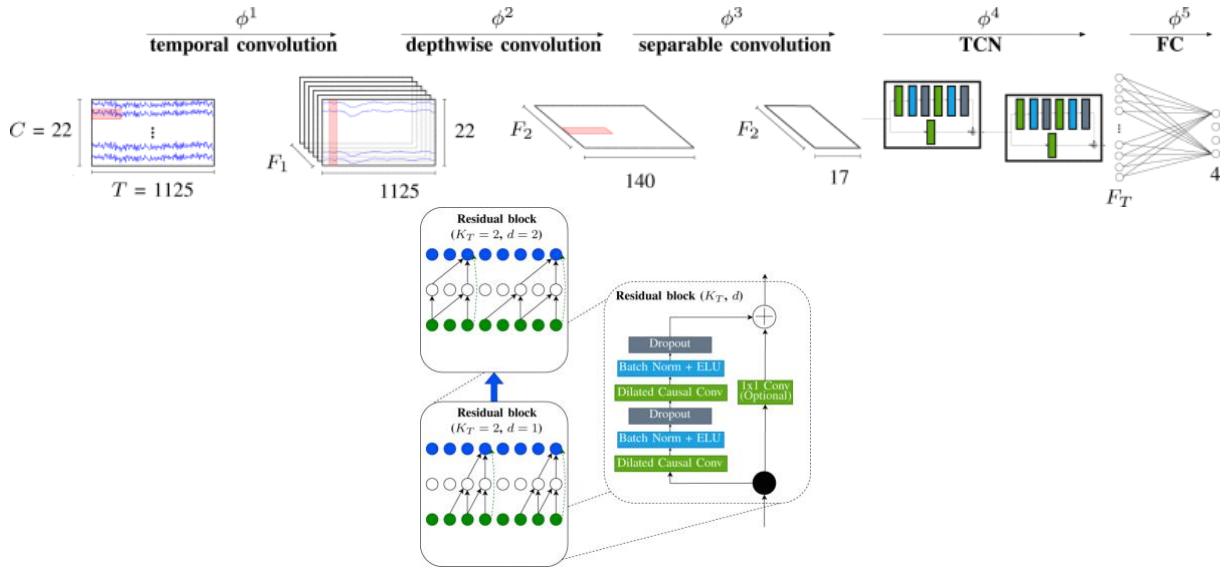

Figure 2: Architecture of the EEG-TCNET on top, with the elements of it's TCN shown below [15].

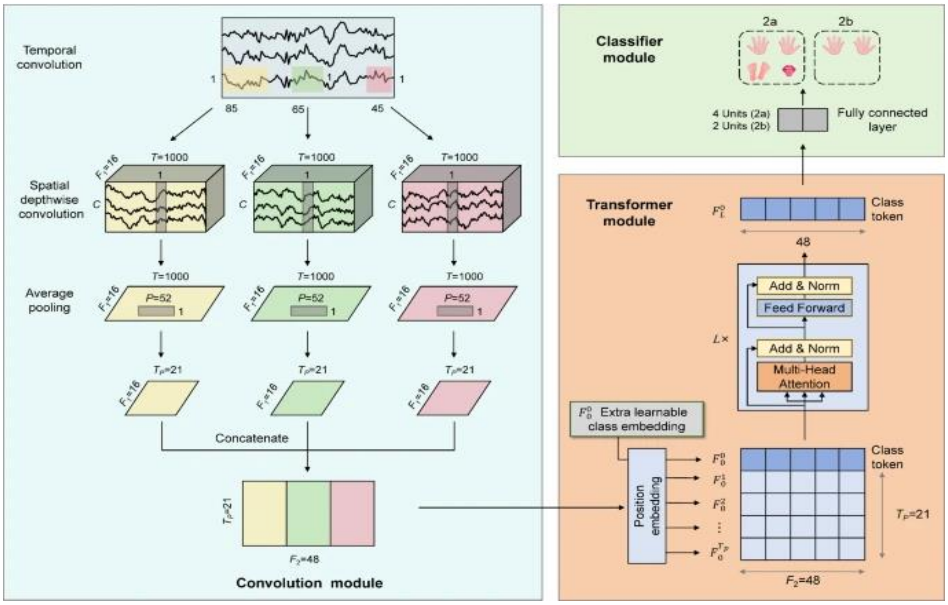

Figure 3: Architecture of MSCFormer, showing the convolution module in blue, the transformer module in orange, and the classifier module in green [16].

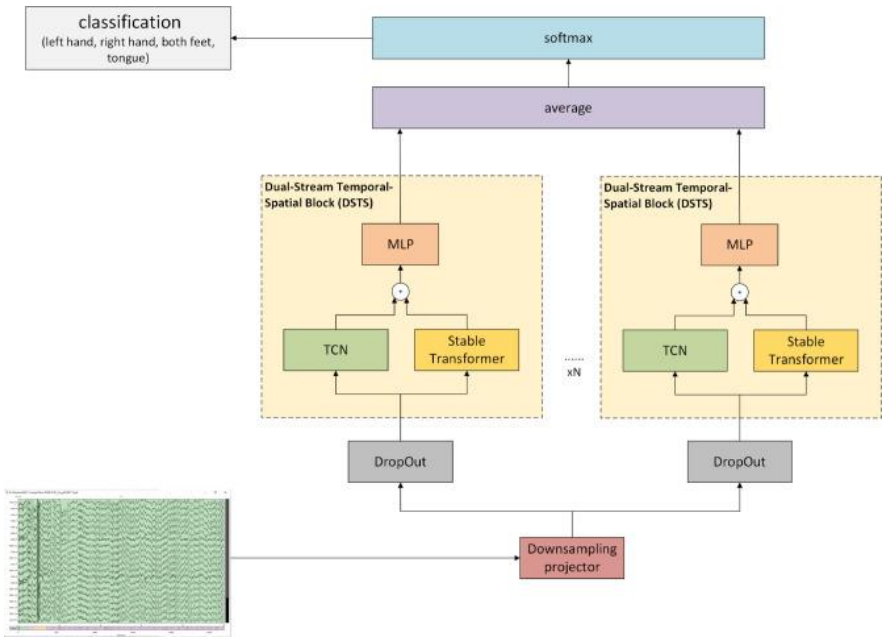

Figure 4: Architecture of the EEGEncoder [17].

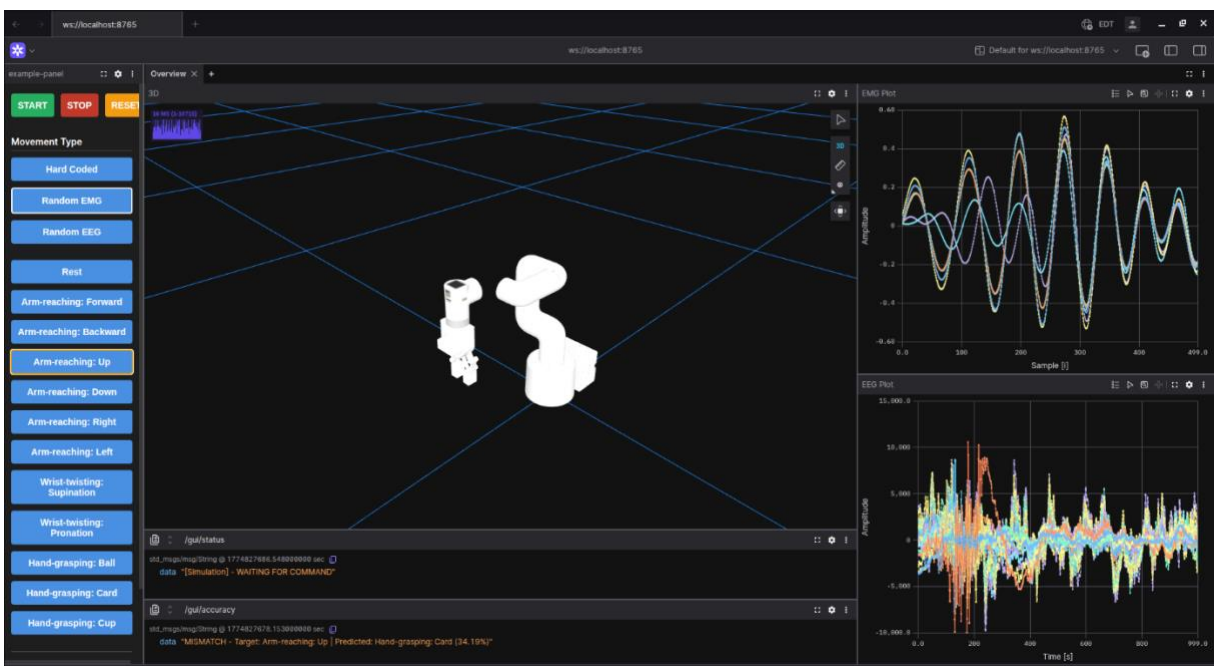

Figure 5: Screenshot of Foxglove GUI used for system evaluation. On the left-hand side is the inference control. The right-hand side shows the raw EMG (above) or raw EEG (below) signals being inferred. The center displays the RViz visualization of the myCobot 280 with the adaptive gripper. Below that is the current state of the robot, either moving or waiting for a command. Further below that is the target and predicted movement, with prediction confidence.

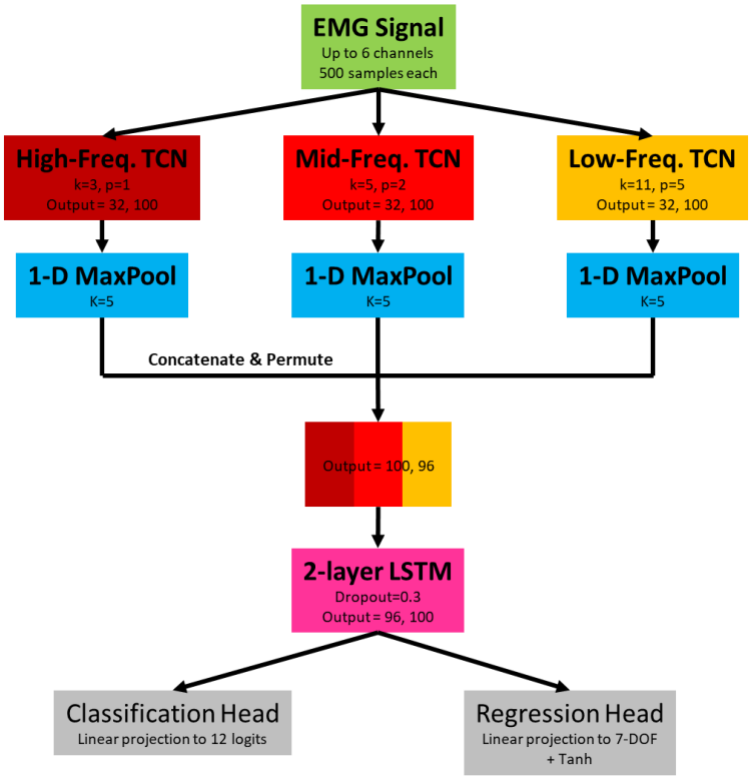

Figure 6: Custom Multi-Head TCN-LSTM architecture.

C. Additional Results

Table 11: Joint angles and gripper values for the 11 movements and rest.

State	Joint 1	Joint 2	Joint 3	Joint 4	Joint 5	Joint 6	Gripper
Rest	0.0	0.0	-90.0	0.0	0.0	0.0	0
Arm-reaching: Forward	0.0	-45.0	-45.0	0.0	0.0	0.0	0
Arm-reaching: Backward	0.0	45.0	135.0	0.0	0.0	0.0	0
Arm-reaching: Up	0.0	20.0	-50.0	0.0	0.0	0.0	0
Arm-reaching: Down	0.0	-20.0	-135.0	0.0	0.0	0.0	0
Arm-reaching: Left	90.0	-45.0	-45.0	0.0	0.0	0.0	0

Arm-reaching: Right	-90.0	-45.0	-45.0	0.0	0.0	0.0	0
Wrist-twisting: Pronation	0.0	0.0	-90.0	-45.0	0.0	0.0	0
Wrist-twisting: Supination	0.0	0.0	-90.0	45.0	0.0	0.0	0
Hand-grasping: Card	0.0	0.0	-90.0	0.0	0.0	0.0	100
Hand-grasping: Ball	0.0	0.0	-90.0	0.0	0.0	0.0	60
Hand-grasping: Cup	0.0	0.0	-90.0	0.0	0.0	0.0	40

Table 12: Evaluated metrics of the four individual subjects across all three EEG models.

Subject	Metric	EEG-TCNet	MSCFormer	EEGEncoder
1 (Male)	Accuracy	70.69%	74.32%	76.43%
	Precision	52.57%	51.44%	57.82%
	Recall	59.51%	57.92%	61.40%
	F1-Score	54.56%	53.22%	58.67%
	Kappa	61.51%	65.54%	68.10%
2 (Female)	Accuracy	70.00%	68.79%	70.91%
	Precision	57.16%	54.44%	55.46%
	Recall	62.75%	60.45%	60.20%
	F1-Score	58.63%	55.54%	56.19%
	Kappa	60.73%	55.54%	61.47%
7 (Male)	Accuracy	67.27%	69.39%	72.73%
	Precision	46.86%	48.18%	50.89%
	Recall	47.45%	53.48%	55.03%
	F1-Score	45.03%	47.09%	52.17%
	Kappa	56.09%	59.51%	62.97%
9 (Female)	Accuracy	75.15%	73.33%	72.12%
	Precision	48.37%	40.89%	50.22%
	Recall	51.20%	44.33%	51.87%
	F1-Score	47.91%	41.27%	50.53%
	Kappa	65.87%	62.43%	62.66%